

1. Questions

Study the following information carefully and answer the questions given below:

A company deals in only two types of chess boards namely wooden and vinyl. The number of wooden boards sold in 2023 was 33.33% more than that in 2024. The number of vinyl boards sold in 2024 were 40% less than that in 2025 and 25% less than the number of wooden boards sold in 2024. The total number of vinyl boards sold in the given three years were twice the number of wooden boards sold in 2025 which was 900 while the total number of chess boards sold during the given three years were 4100.

If the number of vinyl boards sold in 2026 is 32% of the sum of the number of wooden boards sold in 2024 and 2025 together, then find the number of vinyl boards sold in 2026?

- a. 600
- b. 440
- c. 360
- d. 540
- e. 480

2. Questions

If the average of the total number of chess boards (wooden + vinyl + magnetic) sold in 2025 was 700, then find the number of magnetic chess boards sold in 2025?

- a. 476
- b. 484
- c. 450
- d. 320
- e. 396

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3. Questions

Find the ratio of the number of wooden boards sold in 2024 to the number of vinyl boards sold in 2025?

- a. 4:5
- b. 5:3
- c. 4:3
- d. 2:1
- e. 3:1

4. Questions

In 2024, the total number of chess boards sold through online modes are two – thirds of offline

modes. Find the total number of chess boards sold through offline modes in 2024?

- a. 448
- b. 504
- c. 420
- d. 630
- e. 560

5. Questions

The total number of chess boards sold in 2024 were how much percent less than that in 2023?

- a. 33%
- b. 25%
- c. 29%
- d. 20%
- e. 18%

6. Questions

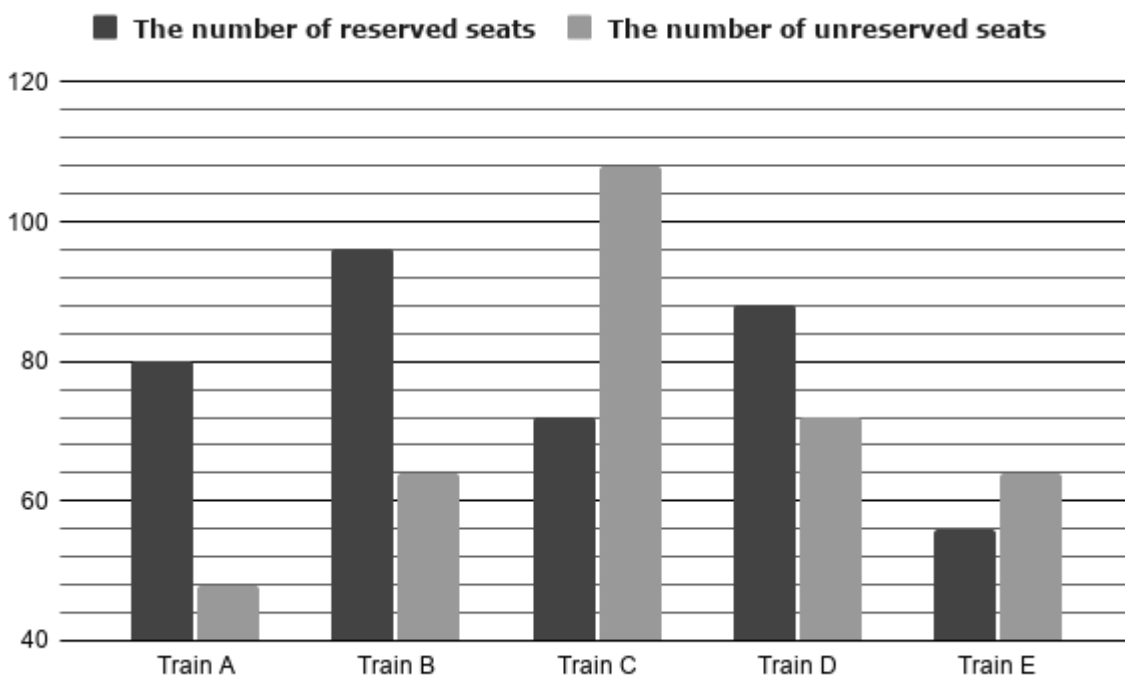
Study the following information carefully and answer the questions given below:

The given bar graph shows the number of reserved seats and the total number of unreserved seats in five different trains namely A, B, C, D and E.

Note:

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For each train, the total number of seats = the number of reserved seats + the number of unreserved seats



The cost of reservation of a seat in train C is Rs. 450. Find the total amount collected by all reserved seats in train C.

- a. Rs. 30800
- b. Rs. 31200
- c. Rs. 32400
- d. Rs. 35600
- e. Rs. 38900

7. Questions

Find the ratio of the number of reserved seats in train B to the sum of the number of unreserved seats in trains A and E together?

- a. 5:6
- b. 8:7
- c. 7:9
- d. 7:5
- e. 6:7

8. Questions

If the difference between the total number of seats in trains A and D is P, then find the value of 1.5P?

- a. 48
- b. 56
- c. 36
- d. 44
- e. 52

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9. Questions

Study the following information carefully and answer the questions given below:

The missing table DI shows the total number of students (boys + girls) in five different schools i.e. P, Q, R, S and T and also given the percentage distribution of the number of girls in the school out of the total number of students.

Schools	The total number of students	The percentage distribution of the number of girls
P	800	-
Q	600	60%
R	-	52%
S	900	-
T	-	68%

If the ratio of the number of boys in school P to that in school Q is 11:6, then find the number of girls in school P?

- a. 380
- b. 360
- c. 375
- d. 324
- e. 356

10. Questions

If the number of girls in school R is 390, then find the ratio of the number of boys in school Q to that in school R?

- a. 3:5
- b. 2:5
- c. 5:4
- d. 2:3
- e. 4:3

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11. Questions

If the number of boys in school T is 160, then find the difference between the total number of students in schools Q and T?

- a. 100
- b. 300
- c. 250
- d. 400
- e. 180

12. Questions

If the average number of girls in schools P and S are 360 and the number of boys in school P is 100 less than that in school S, then find the number of boys in school S?

- a. 520
- b. 360
- c. 540
- d. 440
- e. 390

13. Questions

If the total number of students in school S is 150 more than that in school R, then find the number of girls in school R?

- a. 350
- b. 424
- c. 375
- d. 420
- e. 390

14. Questions

A vessel contains 108 litres mixture of milk and water, in which quantity of milk is 20% less than that of water. 36 litres of mixture is removed from the vessel and 8 litres of water is now added to the vessel. Find the final ratio of the quantity of water to that of milk in the vessel now.

- a. 5:7
- b. 3:2
- c. 7:4
- d. 2:3
- e. 4:7

15. Questions

A certain amount becomes thrice of itself in 8 years at a certain rate of simple interest (per annum). Find the time period in which it will become 11 times of itself at the same rate of simple interest?

- a. 30 years
- b. 60 years
- c. 50 years
- d. 40 years

e. 25 years

16. Questions

A and B started a business with investments of Rs. 900 and Rs. 1200 respectively. After 9 months, 'A' reduces her investment by Rs. 300 whereas 'B' reduces her investment to Rs. 750. If the total profit received at the end of 18 months is Rs. 575, then find the profit share of A.

- a. Rs. 315
- b. Rs. 360
- c. Rs. 250
- d. Rs. 325
- e. Rs. 275

17. Questions

A bike covered 240 km in 6 hours. The ratio of speed of a bike to that of a car is 8:5. If the speed of the car is increased by 20%, then find the distance covered by the car in 4 hours now.

- a. 120 km
- b. 160 km
- c. 180 km
- d. 145 km
- e. 156 km

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18. Questions

The loss incurred on selling 12 articles for Rs. 600 is equal to the cost price of 2 articles. Find the selling price of each article if the seller wants to earn a profit of 20%.

- a. Rs. 75
- b. Rs. 84
- c. Rs. 64
- d. Rs. 88
- e. Rs. 72

19. Questions

P alone can do work in 10 days. P and Q work together to complete the same work in $(35/8)$ days. If both get paid Rs. 5600 in total, then find the share of Q?

- a. Rs. 3600
- b. Rs. 3280

- c. Rs. 3750
- d. Rs. 3150
- e. Rs. 3480

20. Questions

The length of the rectangle is 40% more than the side of the square whose diagonal is $10\sqrt{2}$ cm. If the area of the rectangle is 12 cm^2 more than the area of the square, then find the breadth of the rectangle.

- a. 10 cm
- b. 8 cm
- c. 11 cm
- d. 6 cm
- e. 12 cm

21. Questions

What value should come in the place of (?) in the following questions?

$$24 * 4.5 + 16 * 5.5 = ? \div \sqrt{25}$$

- a. 1020
- b. 960
- c. 980
- d. 1060
- e. 920

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22. Questions

$$513 \div 19 * ? = 16^2 + 68$$

- a. 9
- b. 17
- c. 21
- d. 14
- e. 12

23. Questions

$$\sqrt{?} = \sqrt{196} - \sqrt{256} + \sqrt{121}$$

- a. 121
- b. 25
- c. 9
- d. 81
- e. 64

24. Questions

$$((270 \div 9) \div 2) * ? + 70 = 235$$

- a. 11
- b. 13
- c. 6
- d. 19
- e. 9

25. Questions

$$\sqrt{484} + 15^2 - 22\% \text{ of } 450 = ?$$

- a. 136
- b. 148
- c. 164
- d. 154
- e. 172

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26. Questions

What approximate value should come in the place of (?) in the following questions?

$$(780.09 - ?) \div (144.98 \div 5.01 - 8.12) = 29.98$$

- a. 150
- b. 120
- c. 180
- d. 210
- e. 240

27. Questions

$$(\sqrt{2917} * {}^3\sqrt{510}) \div 26.99 = ?$$

- a. 11
- b. 19
- c. 22
- d. 20
- e. 16

28. Questions

$$(? \div 14.02) \text{ of } \sqrt{195} = (49.23 \div ?) \text{ of } \sqrt{15.98}$$

- a. 23
- b. 17
- c. 14
- d. 9
- e. 21

29. Questions

$$16.01\% \text{ of } 18.23\% \text{ of } 2499.65 + 56.23 \div 2.04 = ?^2$$

- a. 22
- b. 20
- c. 18
- d. 10
- e. 15

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30. Questions

$$1619.88 - (27.04 * 19.89) = ? * 9.09$$

- a. 96
- b. 120
- c. 112
- d. 88
- e. 132

31. Questions

What value should come in the place of (?) in the following number series?

48, 48, 24, 72, 18, ?

- a. 75
- b. 92
- c. 70
- d. 90
- e. 84

32. Questions

16, 21, 31, ?, 74, 111

- a. 51
- b. 42
- c. 48
- d. 54
- e. 45

33. Questions

13, 54, 91, 122, 151, ?

- a. 174
- b. 162
- c. 176
- d. 169
- e. 180

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34. Questions

12, 36, ?, 54, 27, 81

- a. 16
- b. 26
- c. 21
- d. 24
- e. 18

35. Questions

129, 165, ?, 56, 29, 33

- a. 42

- b. 40
- c. 38
- d. 44
- e. 50

36. Questions

Find out the wrong number in the following number series.

129, 146, 163, 180, 195, 214

- a. 214
- b. 129
- c. 163
- d. 195
- e. 146

37. Questions

9, 28, 63, 126, 217, 344

- a. 344
- b. 63
- c. 217
- d. 28
- e. 9

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38. Questions

16, 27, 40, 55, 75, 91

- a. 16
- b. 27
- c. 75
- d. 91
- e. 40

39. Questions

540, 180, 90, 45, 15, 5

- a. 180

- b. 15
- c. 5
- d. 540
- e. 45

40. Questions**18, 27, 43, 64, 88, 112**

- a. 112
- b. 64
- c. 27
- d. 88
- e. 18

41. Questions

The following question contains two equations as I and II. You have to solve both equations and determine the relationship between them and give answer as,

I). $x^2 + 23x + 102 = 0$

II). $y^2 - 8y - 65 = 0$

- a. $x \geq y$
- b. $x \leq y$
- c. $x < y$
- d. $x = y$ or the relation cannot be established.
- e. $x > y$

@RankZenBanker**42. Questions**

I). $x^2 - 23x + 112 = 0$

II). $y^2 + 12y - 133 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x < y$
- d. $x = y$ or the relation cannot be established.
- e. $x \leq y$

43. Questions

I). $x^2 - 169 = 0$

II). $y^3 = 2197$

- a. $x > y$
- b. $x \geq y$
- c. $x \leq y$
- d. $x = y$ or the relation cannot be established.
- e. $x < y$

44. Questions

I). $x^2 - 4x - 96 = 0$

II). $y^2 + 27y + 180 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x < y$
- d. $x = y$ or the relation cannot be established.
- e. $x \leq y$

45. Questions

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I). $x - 30\sqrt{x} + 224 = 0$

II). $y - 27\sqrt{y} + 180 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x \leq y$
- d. $x < y$
- e. $x = y$ or the relation cannot be established.

Explanations:

1. Questions

The total number of vinyl boards sold in the given three years = $900 * 2 = 1800$

Let the number of wooden boards sold in 2023 and 2024 be $4x$ and $3x$ respectively.

The total number of wooden boards sold = $4100 - 1800 = 2300$

According to the question,

$$4x + 3x + 900 = 2300$$

$$7x = 1400$$

$$x = 200$$

The number of wooden boards sold in 2023 = $4x = 4 * 200 = 800$

The number of wooden boards sold in 2024 = $3x = 3 * 200 = 600$

The number of vinyl boards sold in 2024 = $600 * \{1 - (25/100)\}$

$$= 6 * 75 = 450$$

The number of vinyl boards sold in 2025 = $450 * (100/60) = 750$

The number of vinyl boards sold in 2023 = $1800 - (750 + 450) = 600$

Year	The number of wooden boards sold	The number of vinyl boards sold	The total number of chess boards sold
2023	800	600	1400
2024	600	450	1050
2025	900	750	1650
Total	2300	1800	4100

Answer: E

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The sum of the number of wooden boards sold in 2024 and 2025 together = $600 + 900 = 1500$

The number of vinyl boards sold in 2026 = $1500 * (32/100) = 480$

Hence, the correct answer is option E

2. Questions

The total number of vinyl boards sold in the given three years = $900 * 2 = 1800$

Let the number of wooden boards sold in 2023 and 2024 be $4x$ and $3x$ respectively.

The total number of wooden boards sold = $4100 - 1800 = 2300$

According to the question,

$$4x + 3x + 900 = 2300$$

$$7x = 1400$$

$$x = 200$$

The number of wooden boards sold in 2023 = $4x = 4 * 200 = 800$

The number of wooden boards sold in 2024 = $3x = 3 * 200 = 600$

The number of vinyl boards sold in 2024 = $600 * \{1 - (25/100)\}$
 $= 6 * 75 = 450$

The number of vinyl boards sold in 2025 = $450 * (100/60) = 750$

The number of vinyl boards sold in 2023 = $1800 - (750 + 450) = 600$

Year	The number of wooden boards sold	The number of vinyl boards sold	The total number of chess boards sold
2023	800	600	1400
2024	600	450	1050
2025	900	750	1650
Total	2300	1800	4100

Answer: C

The sum of the total number of chess boards (wooden + vinyl + magnetic) sold in 2025 = $700 * 3 = 2100$

The number of magnetic chess boards sold in 2025 = $2100 - (900 + 750) = 450$

Hence, the correct answer is option C

3. Questions

The total number of vinyl boards sold in the given three years = $900 * 2 = 1800$

Let the number of wooden boards sold in 2023 and 2024 be $4x$ and $3x$ respectively.

The total number of wooden boards sold = $4100 - 1800 = 2300$

According to the question,

$$4x + 3x + 900 = 2300$$

$$7x = 1400$$

$$x = 200$$

The number of wooden boards sold in 2023 = $4x = 4 * 200 = 800$

The number of wooden boards sold in 2024 = $3x = 3 * 200 = 600$

The number of vinyl boards sold in 2024 = $600 * \{1 - (25/100)\}$
 $= 6 * 75 = 450$

The number of vinyl boards sold in 2025 = $450 * (100/60) = 750$

The number of vinyl boards sold in 2023 = $1800 - (750 + 450) = 600$

Year	The number of wooden boards sold	The number of vinyl boards sold	The total number of chess boards sold
2023	800	600	1400
2024	600	450	1050
2025	900	750	1650
Total	2300	1800	4100

Answer: A

The required ratio = $600:750 = 4:5$

Hence, the correct answer is option A

4. Questions

The total number of vinyl boards sold in the given three years = $900 * 2 = 1800$

Let the number of wooden boards sold in 2023 and 2024 be $4x$ and $3x$ respectively.

The total number of wooden boards sold = $4100 - 1800 = 2300$

According to the question,

$$4x + 3x + 900 = 2300$$

$$7x = 1400$$

$$x = 200$$

The number of wooden boards sold in 2023 = $4x = 4 * 200 = 800$

The number of wooden boards sold in 2024 = $3x = 3 * 200 = 600$

The number of vinyl boards sold in 2024 = $600 * \{1 - (25/100)\}$
= $6 * 75 = 450$

The number of vinyl boards sold in 2025 = $450 * (100/60) = 750$

The number of vinyl boards sold in 2023 = $1800 - (750 + 450) = 600$

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Year	The number of wooden boards sold	The number of vinyl boards sold	The total number of chess boards sold
2023	800	600	1400
2024	600	450	1050
2025	900	750	1650
Total	2300	1800	4100

Answer: D

In 2024,

Let the total number of chess boards sold through offline modes be $3x$

So, the total number of chess boards sold through online modes = $3x * (2/3) = 2x$

According to the data,

$$3x + 2x = 1050$$

$$5x = 1050$$

$$x = 210$$

The total number of chess boards sold through offline modes = $3 * 210 = 630$

Hence, the correct answer is option D

5. Questions

The total number of vinyl boards sold in the given three years = $900 * 2 = 1800$

Let the number of wooden boards sold in 2023 and 2024 be $4x$ and $3x$ respectively.

The total number of wooden boards sold = $4100 - 1800 = 2300$

According to the question,

$$4x + 3x + 900 = 2300$$

$$7x = 1400$$

$$x = 200$$

The number of wooden boards sold in 2023 = $4x = 4 * 200 = 800$

The number of wooden boards sold in 2024 = $3x = 3 * 200 = 600$

The number of vinyl boards sold in 2024 = $600 * \{1 - (25/100)\}$

$$= 6 * 75 = 450$$

The number of vinyl boards sold in 2025 = $450 * (100/60) = 750$

The number of vinyl boards sold in 2023 = $1800 - (750 + 450) = 600$

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Year	The number of wooden boards sold	The number of vinyl boards sold	The total number of chess boards sold
2023	800	600	1400
2024	600	450	1050
2025	900	750	1650
Total	2300	1800	4100

Answer: B

The required percentage = $\{(1400 - 1050)/1400\} * 100$
 $= \{350/1400\} * 100 = 25\%$

Hence, the correct answer is option B

6. Questions

For train A,

The number of reserved seats = 80

The number of unreserved seats = 48

The total number of seats = $80 + 48 = 128$

Similarly, we can calculate other values.

Trains	The number of reserved seats	The number of unreserved seats	The total number of seats
A	80	48	128
B	96	64	160
C	72	108	180
D	88	72	160
E	56	64	120

Answer: C

The required amount = $450 * 72 = \text{Rs. } 32400$

Hence, the correct answer is option C

7. Questions

For train A,

The number of reserved seats = 80

The number of unreserved seats = 48

The total number of seats = $80 + 48 = 128$

Similarly, we can calculate other values.

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Trains	The number of reserved seats	The number of unreserved seats	The total number of seats
A	80	48	128
B	96	64	160
C	72	108	180
D	88	72	160
E	56	64	120

Answer: E

The sum of the number of unreserved seats in trains A and E together = $48 + 64 = 112$

The required ratio = $96:112 = 6:7$

Hence, the correct answer is option E

8. Questions

For train A,

The number of reserved seats = 80

The number of unreserved seats = 48

The total number of seats = $80 + 48 = 128$

Similarly, we can calculate other values.

Trains	The number of reserved seats	The number of unreserved seats	The total number of seats
A	80	48	128
B	96	64	160
C	72	108	180
D	88	72	160
E	56	64	120

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Answer: A

The difference between the total number of seats in trains A and D = $160 - 128 = 32$

The required value = $1.5 * 32 = 48$

Hence, the correct answer is option A

9. Questions

Answer: B

The number of boys in school Q = $600 * \{1 - (60/100)\} = 600 * (40/100) = 240$

The number of boys in school P = $240 * (11/6) = 440$

The number of girls in school P = $800 - 440 = 360$

Hence, the correct answer is option B

10. Questions

Answer: D

The number of boys in school Q = $600 * \{1 - (60/100)\} = 6 * 40 = 240$

The percentage distribution of the number of boys in school R = $1 - (52/100) = 48\%$

The number of boys in school R = $390 * (48/52) = 360$

The required ratio = $240:360 = 2:3$

Hence, the correct answer is option D

11. Questions

Answer: A

The percentage distribution of the number of boys in school T = $1 - (68/100) = 32\%$

The total number of students in school T = $160 * (100/32) = 500$

The required difference = $600 - 500 = 100$

Hence, the correct answer is option A

12. Questions

Answer: C

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The sum of the number of girls in schools P and S = $360 * 2 = 720$

The sum of the number of boys in schools P and S = $(800 + 900) - 720 = 980$

Let the number of boys in school S be x

So, the number of boys in school P = $x - 100$

According to the data,

$$x - 100 + x = 980$$

$$2x = 1080$$

$$x = 540$$

The number of boys in school S = 540

Hence, the correct answer is option C

13. Questions

Answer: E

The total number of students in school R = $900 - 150 = 750$

The number of girls in school R = $750 * (52/100) = 390$

Hence, the correct answer is option E

14. Questions

Answer: B

Let the quantity of water in the vessel be $5x$ litres

So, the quantity of milk in the vessel = $5x * 80/100 = 4x$ litres

Given that,

$$5x + 4x = 108$$

$$x = 108/9 = 12$$

The quantity of water in the vessel = $5 * 12 = 60$ litres

The quantity of milk in the vessel = $4 * 12 = 48$ litres

The ratio of the quantity of water and milk = $60:48 = 5:4$

The quantity of milk removed = $36 * (4/9) = 16$ litres

The quantity of water removed = $36 - 16 = 20$ litres

Therefore, required ratio = $(60 - 20 + 8):(48 - 16) = 48:32 = 3:2$

Hence, the correct answer is option B

15. Questions

Answer: D

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Let the principal amount be Rs. P

The rate of interest = $R\%$ p.a.

According to the question,

$$3P = P + \{(P * R * 8)/100\}$$

$$2P = (P * R * 8)/100$$

$$R = 25$$

Let the required time be ' T ' years

$$11P = P + \{(P * 25 * T)/100\}$$

$$10P = (P * 25 * T)/100$$

$$T = 40$$

The required time period = 40 years

Hence, the correct answer is option D

16. Questions

Answer: C

The ratio of profit shares of A and B = $\{(900 * 9) + (900 - 300) * 9\} : \{(1200 * 9) + (750 * 9)\}$
 $= (900 + 600) : (1200 + 750)$
 $= 1500 : 1950 = 10 : 13$

The profit share of A = $(10/23) * 575 = \text{Rs. } 250$

Hence, the correct answer is option C

17. Questions

Answer: A

The speed of bike = $240/6 = 40 \text{ km/hr}$

The speed of car = $40 * (5/8) = 25 \text{ km/hr}$

The new speed of car, now = $25 * (120/100) = 30 \text{ km/hr}$

The required distance = $30 * 4 = 120 \text{ km}$

Hence, the correct answer is option A

18. Questions

Answer: E

Let the cost price of each article be Rs. x

$$600 + 2 * x = 12 * x$$

$$10x = 600$$

$$x = 60$$

The cost price of each article = Rs. 60

The selling price of each article when sold at 20% profit = $60 * (120/100) = \text{Rs. } 72$

Hence, the correct answer is option E

19. Questions

Answer: D

Let the total work be 70 units (LCM of 10 and $(35/8)$)

The efficiency of P = $70/10 = 7 \text{ units/day}$

The efficiency of P and Q = $70 \div (35/8) = 16 \text{ units/day}$

So, the efficiency of Q = $16 - 7 = 9 \text{ units/day}$

The required share received by Q = $5600 * (9/16) = \text{Rs. } 3150$

Hence, the correct answer is option D

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20. Questions

Answer: B

The side of the square = diagonal $\div \sqrt{2}$

The side of the square = $10\sqrt{2} \div \sqrt{2} = 10$ cm

The length of rectangle = $10 * (140/100) = 14$ cm

The area of the square = $a^2 = 10^2 = 100$ cm²

The area of the rectangle = $100 + 12 = 112$ cm²

According to the question,

$$l * b = 112$$

$$b = 112 / 14$$

$$b = 8$$

The breadth of the rectangle = 8 cm

Hence, the correct answer is option B

21. Questions

Answer: C

$$24 * 4.5 + 16 * 5.5 = ? \div \sqrt{25}$$

$$108 + 88 = ? \div 5$$

$$196 * 5 = ?$$

$$? = 980$$

Hence, the correct answer is option C

22. Questions

Answer: E

$$513 \div 19 * ? = 16^2 + 68$$

$$27 * ? = 256 + 68$$

$$? = 324 / 27$$

$$? = 12$$

Hence, the correct answer is option E

23. Questions

Answer: D

$$\sqrt{?} = \sqrt{196} - \sqrt{256} + \sqrt{121}$$

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$$\sqrt{?} = 14 - 16 + 11$$

$$\sqrt{?} = 9$$

$$? = 81$$

Hence, the correct answer is option D

24. Questions

Answer: A

$$((270 \div 9) \div 2) * ? + 70 = 235$$

$$(30 \div 2) * ? = 165$$

$$15 * ? = 165$$

$$? = 11$$

Hence, the correct answer is option A

25. Questions

Answer: B

$$\sqrt{484} + 15^2 - 22\% \text{ of } 450 = ?$$

$$22 + 225 - 99 = ?$$

$$? = 148$$

Hence, the correct answer is option B

26. Questions

Answer: A

$$(780.09 - ?) \div (144.98 \div 5.01 - 8.12) = 29.98$$

$$(780 - ?) \div (145 \div 5 - 8) = 30$$

$$(780 - ?) \div (29 - 8) = 30$$

$$780 - ? = 30 * 21$$

$$780 - 630 = ?$$

$$? = 150$$

Hence, the correct answer is option A

27. Questions

Answer: E

$$(\sqrt{2917} * {}^3\sqrt{510}) \div 26.99 = ?$$

$$(\sqrt{2916} * {}^3\sqrt{512}) \div 27 = ?$$

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$$(54 * 8) \div 27 = ?$$

$$? = 16$$

Hence, the correct answer is option E

28. Questions

Answer: C

$$(? \div 14.02) \text{ of } \sqrt{195} = (49.23 \div ?) \text{ of } \sqrt{15.98}$$

$$(? \div 14) \text{ of } \sqrt{196} = (49 \div ?) \text{ of } \sqrt{16}$$

$$(?/14) * 14 = (49/?) * 4$$

$$?^2 = 49 * 4$$

$$? = 7 * 2$$

$$? = 14$$

Hence, the correct answer is option C

29. Questions

Answer: D

$$16.01\% \text{ of } 18.23\% \text{ of } 2499.65 + 56.23 \div 2.04 = ?^2$$

$$16\% \text{ of } 18\% \text{ of } 2500 + 56 \div 2 = ?^2$$

$$72 + 28 = ?^2$$

$$?^2 = 100$$

$$? = 10$$

Hence, the correct answer is option D

30. Questions

Answer: B

$$1619.88 - (27.04 * 19.89) = ? * 9.09$$

$$1620 - (27 * 20) = ? * 9$$

$$1620 - 540 = ? * 9$$

$$? = 1080/9$$

$$? = 120$$

Hence, the correct answer is option B

31. Questions

Answer: D

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The given series follows the following pattern:

$$48 * 1 = 48$$

$$48 \div 2 = 24$$

$$24 * 3 = 72$$

$$72 \div 4 = 18$$

$$18 * 5 = 90$$

Hence, the correct answer is option D

32. Questions

Answer: C

The given series follows the following pattern:

16	21	31	48	74	111
+5	+10	+17	+26	+37	
	+5	+7	+9	+11	

Hence, the correct answer is option C

33. Questions

Answer: A

The given series follows the following pattern:

$$13 + 41 = 54$$

$$54 + 37 = 91$$

$$91 + 31 = 122$$

$$122 + 29 = 151$$

$$151 + 23 = 174$$

Hence, the correct answer is option A

34. Questions

Answer: E

The given series follows the following pattern:

$$12 * 3 = 36$$

$$36 \div 2 = 18$$

$$18 * 3 = 54$$

$$54 \div 2 = 27$$

$$27 * 3 = 81$$

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Hence, the correct answer is option E

35. Questions

Answer: B

The given series follows the following pattern:

$$129 + 6^2 = 165$$

$$165 - 5^3 = 40$$

$$40 + 4^2 = 56$$

$$56 - 3^3 = 29$$

$$29 + 2^2 = 33$$

Hence, the correct answer is option B

36. Questions

Answer: D

The given series follows the following pattern:

$$129 + 17 = 146$$

$$146 + 17 = 163$$

$$163 + 17 = 180$$

$$180 + 17 = 197$$

$$197 + 17 = 214$$

So, **197** should come in place of **195**.

Hence, the correct answer is option D

37. Questions

Answer: B

The given series follows the following pattern:

$$2^3 + 1 = 9$$

$$3^3 + 1 = 28$$

$$4^3 + 1 = 65$$

$$5^3 + 1 = 126$$

$$6^3 + 1 = 217$$

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$$7^3 + 1 = 344$$

So, **65** should come in place of **63**.

Hence, the correct answer is option B

38. Questions

Answer: C

The given series follows the following pattern:

$$16 + 11 = 27$$

$$27 + 13 = 40$$

$$40 + 15 = 55$$

$$55 + 17 = 72$$

$$72 + 19 = 91$$

So, **72** should come in place of **75**.

Hence, the correct answer is option C

39. Questions

Answer: E

The given series follows the following pattern:

$$540 \div 3 = 180$$

$$180 \div 2 = 90$$

$$90 \div 3 = 30$$

$$30 \div 2 = 15$$

$$15 \div 3 = 5$$

So, **30** should come in place of **45**.

Hence, the correct answer is option E

40. Questions

Answer: A

The given series follows the following pattern:

$$18 + (1 * 9) = 27$$

$$27 + (2 * 8) = 43$$

$$43 + (3 * 7) = 64$$

$$64 + (4 * 6) = 88$$

$$88 + (5 * 5) = 113$$

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Another way:

18 27 43 64 88 **113**
9 16 21 24 25
7 5 3 1

So, **113** should come in place of **112**.

Hence, the correct answer is option A

41. Questions

Answer: C

From I,

$$x^2 + 23x + 102 = 0$$

$$x^2 + 6x + 17x + 102 = 0$$

$$x(x + 6) + 17(x + 6) = 0$$

$$(x + 6)(x + 17) = 0$$

$$x = -6, -17$$

From II,

$$y^2 - 8y - 65 = 0$$

$$y^2 + 5y - 13y - 65 = 0$$

$$y(y + 5) - 13(y + 5) = 0$$

$$(y + 5)(y - 13) = 0$$

$$y = -5, +13$$

So, $x < y$

Hence, the correct answer is option C

42. Questions

Answer: B

From I,

$$x^2 - 23x + 112 = 0$$

$$x^2 - 16x - 7x + 112 = 0$$

$$x(x - 16) - 7(x - 16) = 0$$

$$(x - 16)(x - 7) = 0$$

$$x = +16, +7$$

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From II,

$$y^2 + 12y - 133 = 0$$

$$y^2 - 7y + 19y - 133 = 0$$

$$y(y - 7) + 19(y - 7) = 0$$

$$(y - 7)(y + 19) = 0$$

$$y = +7, -19$$

So, $x \geq y$

Hence, the correct answer is option B

43. Questions

Answer: C

From I,

$$x^2 - 169 = 0$$

$$x = +13, -13$$

From II,

$$y^3 = 2197$$

$$y = +13$$

So, $x \leq y$

Hence, the correct answer is option C

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44. Questions

Answer: A

From I,

$$x^2 - 4x - 96 = 0$$

$$x^2 - 12x + 8x - 96 = 0$$

$$x(x - 12) + 8(x - 12) = 0$$

$$(x - 12)(x + 8) = 0$$

$$x = +12, x = -8$$

From II,

$$y^2 + 27y + 180 = 0$$

$$y^2 + 12y + 15y + 180 = 0$$

$$y(y + 12) + 15(y + 12) = 0$$

$$(y + 12)(y + 15) = 0$$

$$y = -12, -15$$

So, $x > y$

Hence, the correct answer is option A

45. Questions

Answer: E

From I,

$$x - 30\sqrt{x} + 224 = 0$$

$$x - 14\sqrt{x} - 16\sqrt{x} + 224 = 0$$

$$\sqrt{x}(\sqrt{x} - 14) - 16(\sqrt{x} - 14) = 0$$

$$(\sqrt{x} - 14)(\sqrt{x} - 16) = 0$$

$$\sqrt{x} = 14, 16$$

$$x = 196, 256$$

From II,

$$y - 27\sqrt{y} + 180 = 0$$

$$y - 15\sqrt{y} - 12\sqrt{y} + 180 = 0$$

$$\sqrt{y}(\sqrt{y} - 15) - 12(\sqrt{y} - 15) = 0$$

$$(\sqrt{y} - 15)(\sqrt{y} - 12) = 0$$

$$\sqrt{y} = 15, 12$$

$$y = 225, 144$$

So, the relationship between x and y cannot be established.

Hence, the correct answer is option E

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